EXPERIMENT 18

MATLAB SIMULATION AND EVALUATION OF EYE DIAGRAM PARAMETERS UNDER INTER-SYMBOL INTERFERENCE IN DIGITAL COMMUNICATION SYSTEMS

OBJECTIVE:

1. To simulate a digital communication system in MATLAB and generate the eye diagram of the received signal for evaluating eye height, eye width, eye-crossing level and timing margin.

2. To introduce and study Inter-Symbol Interference (ISI) under different channel conditions, and evaluate its effect on eye-diagram opening and overall digital signal quality.

PROGRAM:

```
clc;

clear;

close all;

%% Parameters

N = 100;       % Number of bits

sps = 20;      % Samples per symbol

%% Generate Random Binary Data

data = randi([0 1],1,N);

%% NRZ Encoding

tx = repelem(data,sps);

%% Time Axis

t = (0:length(tx)-1)/sps;

%% Introduce Inter-Symbol Interference (ISI)

% Simple Low-Pass FIR Filter (Channel)

h = [0.2 0.6 1 0.6 0.2];


% Received Signal with ISI
```

```
rx = conv(tx,h,'same');
%% Plot Results
figure;
subplot(4,1,1)
stairs(data,'LineWidth',1.5)
grid on
title('Original Binary Data')
xlabel('Bit Number')
ylabel('Bit')
ylim([-0.2 1.2])
subplot(4,1,2)
plot(t,tx,'LineWidth',1.5)
grid on
title('Transmitted NRZ Signal')
xlabel('Time')
ylabel('Amplitude')
ylim([-0.5 1.5])
subplot(4,1,3)
plot(t,rx,'r','LineWidth',1.2)
grid on
title('Received Signal with Inter-Symbol Interference (ISI)')
xlabel('Time')
ylabel('Amplitude')

subplot(4,1,4)
hold on
% Manual Eye Diagram
for i = 1:2*sps:length(rx)-2*sps
```

```
    segment = rx(i:i+2*sps-1);
    plot((0:2*sps-1)/sps,segment,'b')
end
grid on
title('Eye Diagram under ISI')
xlabel('Time (Bit Periods)')
ylabel('Amplitude')
hold off
```

OUTPUT:

